b) 100w capacity of 40 lamps glow 8 hours per day

$$\text{Energy} = \frac{100 \times 40 \times 8}{1000} = \frac{32000}{1000}\text{Wh}$$
$$= 32\text{kWh} = 32 \text{ units per day}$$

c) 1500w heater works 6 hours per day

$$\text{Energy} = \frac{1500 \times 6}{1000} = \frac{9000}{1000}\text{Wh}$$
$$= 9\text{kWh} = 9 \text{ units per day}$$

The total number of units consumed in 30 days

$$= (11.190 + 32 + 9) \times 30$$
$$= 52.19 \times 30$$
$$= 1565.7 \text{ units}$$

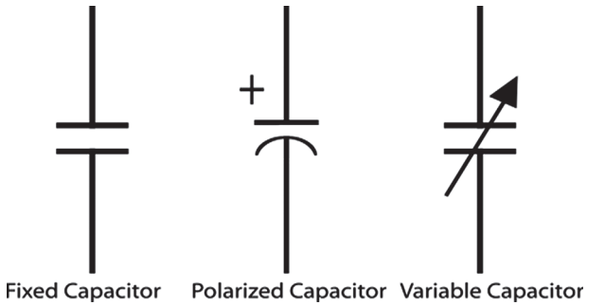
Cost of electricity by for 30 days (1 unit=Rs.6)

$$= 1565.7 \times 6$$
$$= 9394.2$$
$$\text{Cost} = \text{Rs.}9394$$

2.7 CAPACITORS

Capacitor can be defined as, two electrodes are separated by an insulating di-electric medium. It is a device to store electrical energy and to release it when required.

The charge in the capacitor is denoted by capacitance (Q) and the unit is farad (F). The Di-electric medium can be air, mica, wax coated paper or oil etc.



Symbols of Capacitor

2.7.1 Working Principle

In Fig 2.9, a simple parallel plate capacitor is connected with a battery.

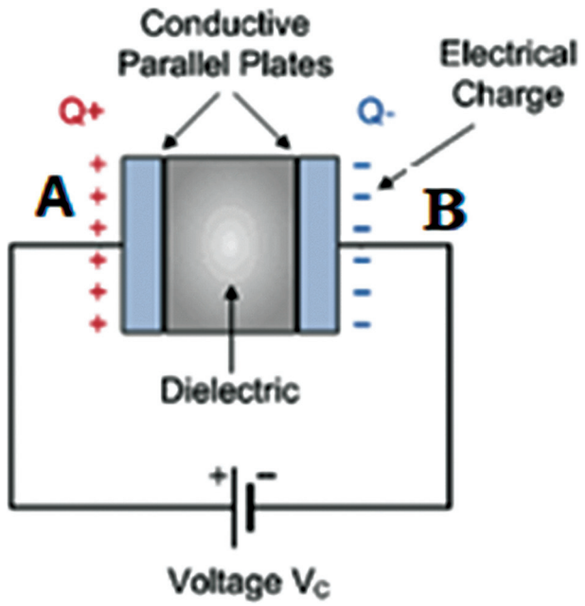


Fig 2.9 *Parallel Plate Capacitor*

Suppose plate 'A' is connected to the +ve terminal and plate 'B' is connected to –ve terminal of DC supply. On closing the switch there will be momentary flow of electrons from positive to negative. Some electrons are withdrawn from the plate 'A', leaving positively charged and transferred to plate 'B' giving it a negative charge. This flow of electrons gives charging current which decreases and finally ceases when the voltage across the capacitor plate has

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